BUAN 5310 – Group Project Report

Modelling Passengers' Airport and Airline Choice Behavior

Seoul Metropolitan Area • Incheon ICN vs. Gimpo GMP

Group Members

Talent Jangano

Shalyne Kirui

Kassim Abdulai

Ednah Nyabengi

March 2026

Contents

1. Introduction 1

2. Exploratory Data Analysis 2

2.1 Dataset Overview 2

2.2 Choice Variable Distributions 3

2.3 Behavioral Choice Patterns 4

2.4 Access Time and Airfare 5

2.5 Missing Values and Data Quality 5

2.6 Mode of Transport to Airport 6

2.7 Variable Transformation 6

3. Models Description 7

4. Model Implementation 7

4.1 Model 1: Airport Choice — Logistic Regression 7

4.3 Model 2: Airport Choice — Decision Tree 10

4.4 Model 3: Airline Choice — Logistic Regression 11

4.5 Model 4: Airline Choice — Decision Tree 12

4.6 Integrated Behavioral Summary 14

4.7 Model Evaluation 15

5. Conclusions and Recommendations 16

References 20

1. Introduction

Airports and airlines compete for passengers on factors beyond proximity and price. In multi-airport regions (MARs), passengers make complex trade-offs, weighing ground access time against flight frequency, trip purpose against airfare sensitivity, and demographic factors against service quality. The central challenge of airport and airline choice research is to quantify these trade-offs in an interpretable manner.

This study examines passenger choice behavior in the Seoul Metropolitan Area, which is served by two competing international airports: Incheon International Airport (ICN) and Gimpo International Airport (GMP). ICN, the larger hub, opened in 2001 and is located approximately 52 kilometers from central Seoul, offering broad global route coverage and high-capacity facilities. GMP is closer to the city center (approximately 18 kilometers) and primarily handles domestic routes and limited short-haul international services to Japan and China. Survey results show a near-even split between ICN (49%) and GMP (51%), with 239 and 249 passengers respectively, confirming that passengers are making genuine choices.

A related behavioral question concerns airline preference. Four carrier groups operate in this market: Korean Air (KE), Asiana Airlines (OZ), Korean low cost carriers (LCCs), and foreign carriers primarily based in China, Japan, and Southeast Asia. Korean carriers collectively account for approximately 71% of the sample. Characterizing which passenger types choose Korean versus foreign carriers elucidates the behavioral structure of airline competition in this market.

The purpose of this project is descriptive analysis. We develop models whose outputs coefficients, odds ratios, and decision rules describe passenger behavior, rather than focusing on maximizing the accuracy of choice replication. We selected logistic regression as the primary descriptive tool because its odds ratios translate directly into behavioral statements, such as: "Passengers traveling for leisure are X times more likely to choose GMP than business travelers." Decision trees complement this approach by generating explicit, human readable decision rules that map the hierarchy of behavioral factors, revealing the most important variable, the second most important, and how they interact to produce observable choice patterns. Both models serve the same purpose: to make passenger behavior readily understandable.

This study builds upon the established literature on airport and airline choice. Leon (2011) used binary logistic regression on a two airport US corridor and found that trip purpose was the dominant behavioral driver, with business travelers significantly more likely to use the local airport due to their preference for connectivity over proximity. Loo (2008) conducted a stated preference study at Hong Kong International Airport and identified airfare, access time, flight frequency, and number of airlines as the four primary level of service attributes shaping airport choice in an Asian MAR a context closely analogous to Seoul. Hess and Polak (2006) analyzed London's five airport system and confirmed that access time and cost are significant behavioral predictors, with business travelers exhibiting heightened sensitivity to access time. Drabas and Wu (2013) extended this framework to airline choice, demonstrating that passengers can be meaningfully segmented by their sensitivity to price, schedule, and carrier loyalty, and that ignoring this heterogeneity produces misleading models. Collectively, these studies inform the behavioral mechanisms that this project seeks to describe in the Seoul market.

2. Exploratory Data Analysis

2.1 Dataset Overview

The dataset contains 488 passenger survey responses collected at Incheon and Gimpo airports. It includes 28 variables across four categories: (1) choice outcomes airport and airline selected; (2) sociodemographic characteristics age, gender, nationality, occupation, income; (3) trip characteristics purpose, duration, destination, seat class, airfare, group travel status, and prior travel frequency; and (4) access journey attributes transport mode, access time, and access cost. No duplicate records were found. Table 1 summarizes the key sample characteristics.

Variable	Category / Statistic	Count	Percent
Age	Mean 40.0 yrs, range 17–80	487	99.8%
Gender	Male / Female	229 / 256	47% / 53%
Trip Purpose	Business	324	66.4%
	Leisure	106	21.7%
	Study / Other	58	11.9%
Seat Class	Economy	437	89.5%

Variable	Category / Statistic	Count	Percent
	Business/First (Premium)	47	9.6%
Destination	SE Asia / Japan / China / Other	170/156/133/24	35/32/27/5%
Access Time	Mean 51.8 min (range 4–390)	391 obs.	80.1%
Airfare	Mean 504,900 KRW (range 30K–2.6M)	333 obs.	68.2%

Table 1. Key sample characteristics (N = 488). Access time and airfare counts reflect non-missing observations.

2.2 Choice Variable Distributions

Figure 1 shows the distributions of airport and airline choices. Airport choice is nearly balanced, with GMP slightly ahead (51% vs. 49%). This relative equivalence is a defining feature of the Seoul MAR: it means both airports are genuinely competitive and that behavioral variables, not just geography, determine which gateway passengers use. Airline choice is more concentrated: Korean carriers collectively account for approximately 71% of the sample, with Korean Air alone holding the largest single share (31.4%). Foreign carriers, primarily from China, Japan, and Southeast Asia, represent 28.7%.

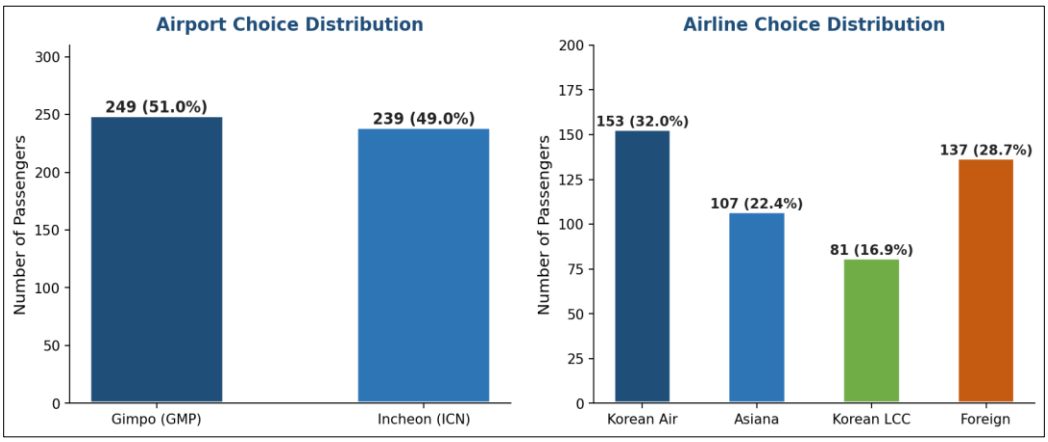

Figure 1. Airport choice distribution (left) and airline choice distribution (right). N=488 for airport; N=478 for airline (10 missing).

2.3 Behavioral Choice Patterns

Before modeling, cross-tabulation reveals preliminary behavioral patterns that the models will later formalize and quantify. Figure 2 breaks down airport choice by trip purpose and age group.

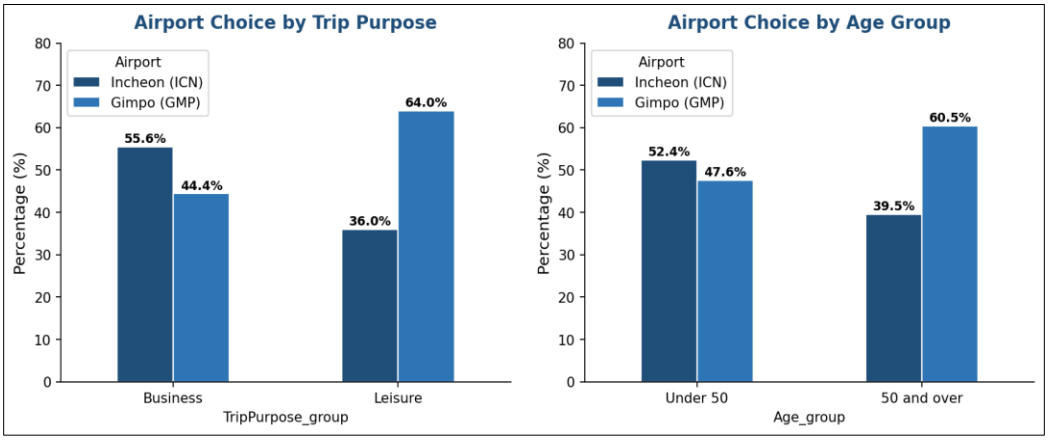

Figure 2. Airport choice by trip purpose (left) and age group (right). These patterns establish the behavioral hypotheses the logistic models test.

Even without a predictive model, clear behavioral patterns emerge. Business travelers prefer ICN (55.6%) over GMP (44.4%), a 10-percentage-point difference favoring the larger hub. However, this preference reverses sharply among leisure travelers, with 64.0% choosing GMP. This 20-point shift between business and leisure travelers is a key behavioral indicator. Age exhibits a similar trend: passengers under 50 are nearly evenly split between GMP (52.4%) and ICN (47.6%), while those 50 and over show a distinct preference for GMP (60.5% vs. 39.5%).

Destination also influences airline choice. Passengers traveling to Southeast Asia choose Korean carriers at a rate of 83.5%, while those flying to China (65.4%) and Japan (66.0%) also show a strong preference for Korean carriers, although at a slightly lower rate. This likely reflects the more extensive route network Korean carriers offer to Southeast Asian destinations compared to their competitors in Northeast Asia.

2.4 Access Time and Airfare

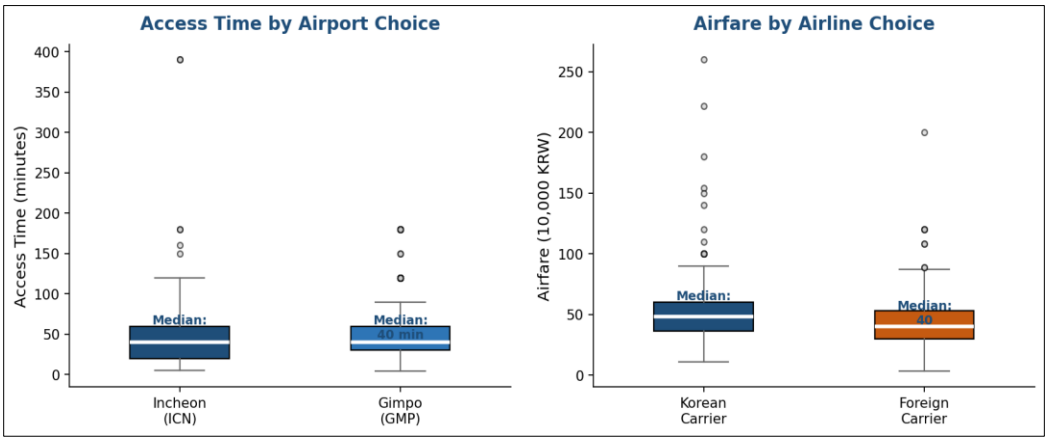

Figure 3. Access time by airport choice (left) and airfare by airline choice (right). ICN passengers have a 10-minute higher median access time, consistent with greater distance from central Seoul.

ICN passengers have a median access time of 50 minutes, compared to 40 minutes for GMP passengers, a difference that reflects ICN's greater distance from the Seoul city center. The overlap in distributions, however, shows that many GMP passengers still have substantial travel times, indicating that not all GMP passengers have a proximity advantage. Airfare distributions between Korean and foreign carriers are similar, with Korean carriers showing a slightly higher median (400,000 vs. 360,000 KRW), consistent with their fuller service positioning relative to Asian low cost carriers.

2.5 Missing Values and Data Quality

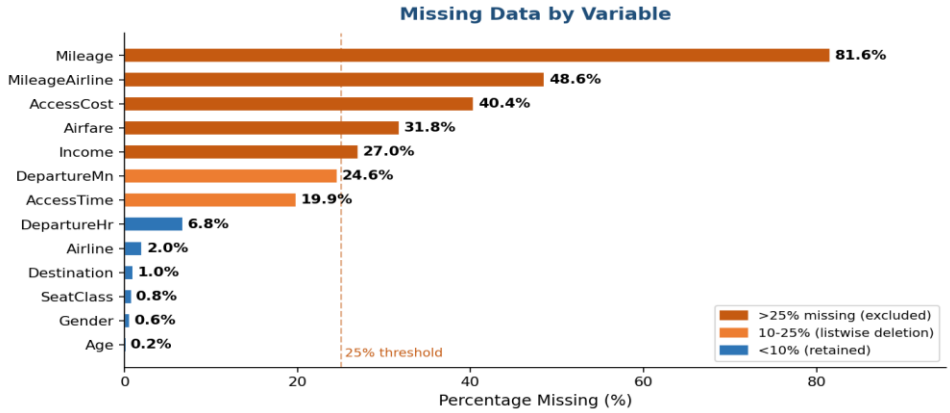

Figure 4. Missing data rates. Variables above 25% were excluded from models. Variables with 10–25% missing were handled via complete-case analysis (listwise deletion).

Three variables were left out of the models because they had too much missing data. Mileage (81.6%) and MileageAirline (48.6%) were missing for too many observations to be useful, and Income (27.1%) was also excluded because dropping all incomplete cases would have reduced the sample too much. Important access related variables AccessCost (40.4%), Airfare (31.8%), and AccessTime (19.9%) were kept in the analysis, but because they also had missing values, the final sample size fell to 259 for the airport model and 326 for the airline model.

2.6 Mode of Transport to Airport

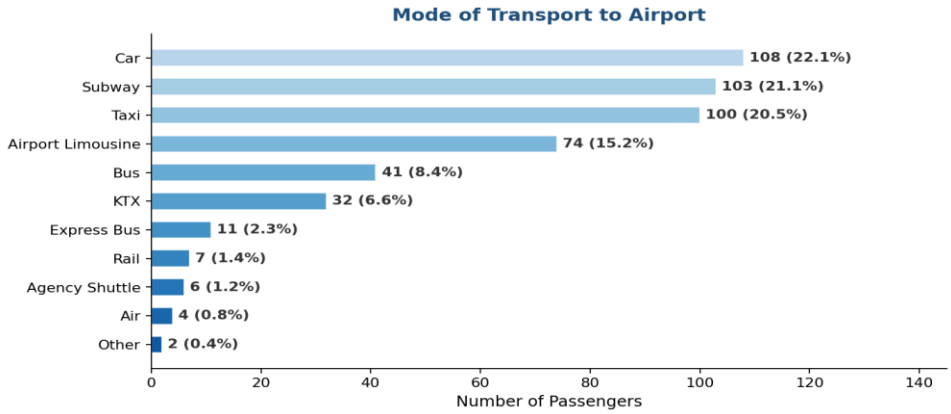

Figure 5. Mode of transport to airport. Private car (22.1%), subway (21.1%), and taxi (20.5%) are the three most common access modes, together accounting for 64% of all journeys.

The multimodal access structure reflects Seoul's well developed public transit network. Airport Limousine buses account for 15.2% of journeys, a mode particularly associated with ICN access. KTX and rail together account for 8% of access trips, with the AREX express rail connection being an important advantage for ICN that is not available for GMP. This transport profile contextualizes why access time and cost may influence airport choice differently across passenger segments.

2.7 Variable Transformation

Original Variable	Transformed Variable	Transformation Logic
TripPurpose (4 cats.)	TripPurpose_group	Business (code 1) vs. Leisure (all others) — follows Leon (2011)
Age (continuous)	Age_group	Under 50 vs. 50 and over
SeatClass (3 cats.)	SeatClass_group	Economy vs. Premium (Business+First merged; only 12 First Class)
Gender (numeric)	Gender_group	Male (1) vs. Female (2)
Airline (4 cats.)	airline_korean	Korean carrier (KE+OZ+LCC = 1) vs. Foreign carrier (= 0)

Table 2. Variable transformation decisions. The business leisure classification follows Leon (2011), while the airline binary transformation follows project recommendations.

3. Models Description

Two descriptive model types are applied to each of the two choice outcomes, giving four models in total. Binary logistic regression serves as the primary descriptive model because it is appropriate for binary choice outcomes and estimates how each predictor changes the odds of selecting one option rather than the other, holding other factors constant. Its coefficients, expressed as odds ratios, directly quantify the behavioral effect of each variable on choice in a way that is interpretable by airport planners and airline managers. The decision tree serves as a complementary descriptive tool by recursively splitting passengers into subgroups based on the most informative predictors. This produces a visual hierarchy of decision rules that shows which factors matter most, in what order they matter, and how they interact to shape observable choice patterns.

The focus throughout is on describing passenger behavior. Model evaluation metrics, including accuracy and ROC AUC, are reported to provide context for the strength of the behavioral signal in the data, but they are not the primary output. The primary outputs are the odds ratios, coefficients interpretations, and the decision tree rules.

4. Model Implementation

4.1 Model 1: Airport Choice — Logistic Regression

4.2.1 Model Specification

The airport choice logistic regression models the probability that a passenger departs from Incheon ICN (coded 1) versus Gimpo GMP (coded 0). After listwise deletion, the modeling dataset contained 259 observations. The model included Age, AccessTime, AccessCost, Gender, TripPurpose, and SeatClass as predictors.

4.2.2 Results

Variable	Coefficient	Std. Error	Odds Ratio	p-Value	Sig.
Intercept	1.318	0.452	3.734	0.004	**
Trip Purpose [Leisure vs. Business]	−0.931	0.282	0.394	0.001	***
Age (per additional year)	−0.025	0.010	0.976	0.017	*
Gender [Male vs. Female]	0.128	0.266	1.136	0.631	n.s.
Seat Class [Premium vs. Economy]	−0.177	0.441	0.838	0.689	n.s.

Variable	Coefficient	Std. Error	Odds Ratio	p-Value	Sig.
Access Time (per minute)	−0.000	0.003	1.000	0.967	n.s.
Access Cost (per KRW)	0.000	0.000	1.000	0.668	n.s.

*Table 3. Airport logistic regression results. The model was estimated in Python using the statsmodels logit function. Reference categories: Business travel, Female, and Economy class. N = 259. Pseudo R^2 = 0.051. LLR p value = 0.006. ***p < 0.001, *p < 0.01, p < 0.05, n.s. = not significant.*

4.2.3 Interpretation of Significant Variables

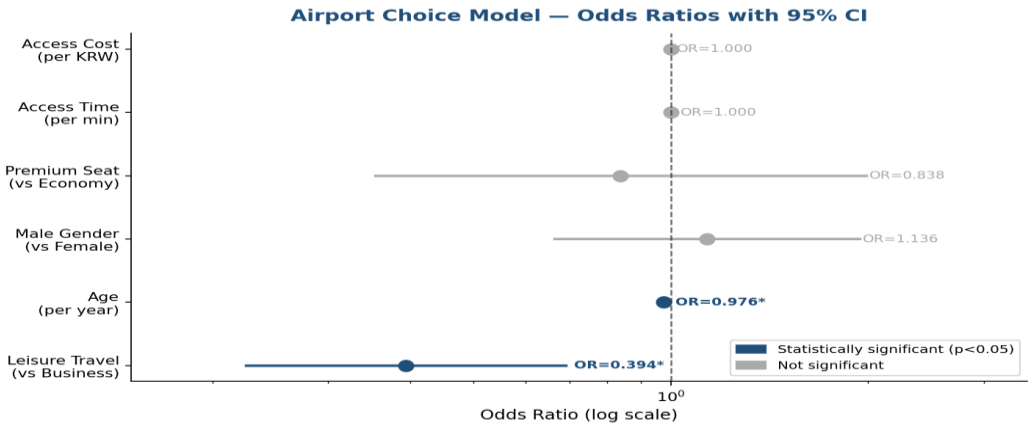

Figure 6. Odds ratios with 95% confidence intervals for the airport logistic model. Only Trip Purpose and Age reach statistical significance. The dashed vertical line at OR=1.0 represents no effect.

Trip Purpose is the dominant driver of airport choice. Leisure travelers have an odds ratio of 0.394 relative to business travelers. Alternatively, a leisure traveler is 2.54 times more likely to depart from Gimpo (GMP) than a business traveler, holding all other factors constant. Stated differently, business travelers are 2.54 times more likely than leisure travelers to choose Incheon (ICN). This is the single most important behavioral finding of this study. It tells us that the Seoul airport market is behaviorally segmented along trip purpose: ICN is effectively the business traveler's airport, and GMP is effectively the leisure traveler's airport.

This behavioral pattern is consistent with the mechanism identified by Leon (2011), who found business travelers 5.21 times more likely to use their local airport in the Fargo Minneapolis corridor because they prioritized connectivity and on time performance over ground access cost. In the Seoul context, the equivalent logic applies: ICN offers a much broader international route network and higher flight frequency, which business travelers value because it gives them more schedule options, fewer connections, and better rebooking flexibility in case of delays. GMP's comparative

advantage, proximity and lower access time, is more strongly valued by leisure travelers who are more sensitive to ground travel burden and cost.

Age is the second significant variable. The odds ratio of 0.976 per additional year of age means: For each additional year of age, a passenger's odds of choosing ICN decline by approximately 2.4%. For example, a 60-year-old passenger's odds of choosing ICN are approximately 26% lower than those of a 35-year-old passenger, all else being equal. Passengers 50 and over are meaningfully more likely to choose GMP than younger passengers.

This behavioral pattern likely reflects several intersecting mechanisms. GMP was Seoul's primary international gateway for decades before ICN opened in 2001, meaning older travelers may have strong habitual familiarity with GMP. Older passengers may also prefer GMP's smaller, less complex terminal environment, a preference consistent with Loo's (2008) finding that non business passengers in Hong Kong placed higher value on simpler, less congested access. Additionally, older travelers may be more sensitive to the extra 10 minutes of median access time that ICN requires.

Access time, access cost, gender, and seat class were not significant predictors of airport choice in this dataset. That result is still useful to note. In particular, prior studies have consistently identified access time as an important factor. Hess and Polak (2006) found it to be the strongest predictor in London, while Loo (2008) found it to be the most statistically important variable in Hong Kong. In this study, the lack of significance is most likely due to missing data: AccessTime was missing for 97 respondents (19.9%) and AccessCost for 197 respondents (40.4%), which reduced the usable sample from 488 to 259. Because earlier research strongly supports the importance of access time, this result should be interpreted as a limitation of the data rather than evidence that access time does not matter in Seoul.

4.3 Model 2: Airport Choice — Decision Tree

4.3.1 What the Tree Tells Us About Passenger Behavior

The airport decision tree (maximum depth = 3) describes passenger behavior through a hierarchy of decision rules. The tree's splits are not performance metrics. They are behavioral descriptions of how the population naturally divides. Figure 7 shows the relative importance of each variable in constructing these rules.

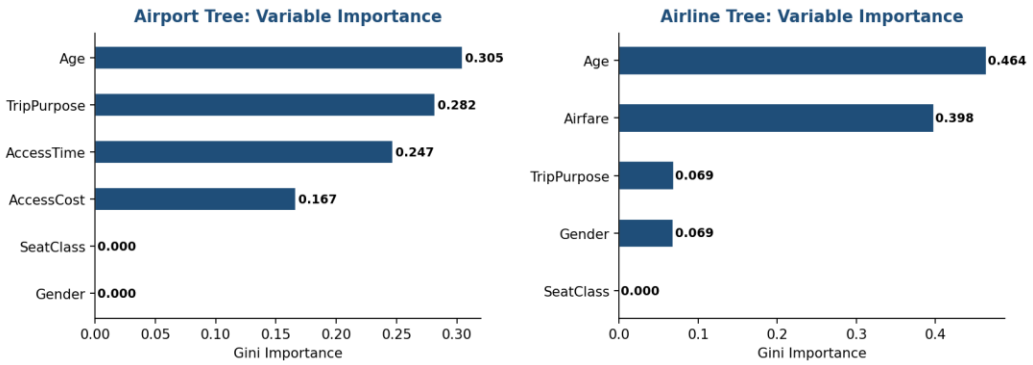

Figure 7. Variable importance from the airport (left) and airline (right) decision trees. Importance is measured by the Gini impurity reduction each variable contributes. Higher values mean the variable is more central to describing the behavioral split.

The airport decision tree generates the following behavioral rules:

Rule 1 Age is the first behavioral split. Passengers aged 55 and under are separated from those over 55. This is the single most discriminating variable in the tree (Gini importance = 0.305), consistent with the logistic regression finding that age significantly predicts airport choice.

Rule 2 Among passengers aged 55 and under, trip purpose determines airport choice. Passengers under 56 who are traveling for business are classified as ICN passengers. Passengers under 56 traveling for leisure are classified as GMP passengers regardless of their access time or cost. This confirms that among the majority of passengers, the business leisure split is the decisive behavioral dividing line.

Rule 3 Among passengers over 55, access time becomes the key behavioral factor. Passengers over 55 with access times of 17.5 minutes or less tend toward GMP; those with access times over 17.5 minutes also tend toward GMP, but with variation based on access cost. This rule describes a behavioral pattern in which older passengers are more sensitive to the ground access dimension.

Their airport choice is not determined solely by trip purpose but is also shaped by how long it takes them to get to the airport.

Taken together, the airport decision tree describes a Seoul market with a clear two tier behavioral structure: age first divides the population into two groups with different behavioral profiles, and within the younger majority, trip purpose is the decisive factor. This nested structure provides a richer description of behavior than the logistic regression coefficients alone because it shows that the effect of trip purpose is primarily relevant for passengers under 55, while older passengers follow a different behavioral logic centered on access convenience.

4.4 Model 3: Airline Choice — Logistic Regression

4.4.1 Results

Variable	Coefficient	Std. Error	Odds Ratio	p-Value	Sig.
Intercept	0.666	0.459	1.946	0.147	n.s.
Gender [Male vs. Female]	−0.331	0.251	0.718	0.188	n.s.
Trip Purpose [Leisure vs. Business]	0.033	0.266	1.034	0.900	n.s.
Seat Class [Premium vs. Economy]	−0.301	0.421	0.740	0.474	n.s.
Age (per additional year)	−0.000	0.009	1.000	0.966	n.s.
Airfare (per 10,000 KRW)	0.007	0.005	1.007	0.137	n.s.

Table 4. Airline logistic regression results. Reference categories: Business travel, Female, Economy. N=326. Pseudo R^2=0.011. LLR p-value=0.509. n.s.=not significant.

4.4.2 What the Airline Logistic Model Tells Us

The results of the airline logistic regression indicate that none of the explanatory variables included in the model are statistically significant predictors of airline choice. In particular, the overall model likelihood ratio test (LLR) p-value of 0.509 suggests that the set of variables considered does not explain airline selection better than random chance. Rather than indicating a weakness in the modeling approach, this outcome provides a meaningful behavioral insight.

Specifically, it implies that the demographic and trip-related characteristics collected in the survey such as age, gender, travel purpose, seat class, and airfare do not significantly differentiate passengers who choose Korean carriers from those who select foreign airlines.

From a behavioral perspective, this finding suggests that airline selection in this market is less influenced by passenger demographics and more likely determined by factors not captured in the available dataset. Prior research supports this interpretation. For instance, Tomasz Drabas and Chunyan Wu (2013) demonstrated that brand loyalty, particularly whether a passenger flew with the same airline on their previous trip is one of the strongest predictors of subsequent airline choice, often outweighing demographic characteristics. Similarly, Becky P. Y. Loo (2008) found that route availability, including the number of airlines serving a route and flight frequency, plays a critical role in shaping both airport and airline preferences. Importantly, the current survey dataset does not include variables related to frequent flyer program membership, prior airline usage, or route-level service characteristics. The absence of these behavioral and service-related variables likely explains why demographic and trip attributes alone do not significantly account for airline choice in this analysis.

Although the regression coefficients are not statistically significant, their directional patterns still offer limited behavioral interpretation. The results suggest that male passengers exhibit a slight tendency toward foreign carriers (odds ratio = 0.718). This pattern may reflect broader international travel trends in which male travelers, particularly those traveling for business, may have greater exposure to non-Korean carriers operating within Asian and intercontinental markets. Additionally, airfare shows a small positive association with choosing a Korean carrier (odds ratio = 1.007 per 10,000 KRW). This trend is consistent with the market positioning of major Korean airlines, such as Korean Air and Asiana Airlines, which traditionally operate as full-service carriers and often command higher fares compared with some foreign low-cost or regional competitors. While these patterns are not statistically significant, they remain consistent with established industry dynamics and provide useful contextual insight for interpreting airline choice behavior.

4.5 Model 4: Airline Choice — Decision Tree

4.5.1 What the Tree Tells Us About Passenger Behavior

The airline decision tree uses Airfare and Age as its primary splitting variables. Figure 7 shows that Age accounts for 46.4% of the tree's total Gini importance and Airfare accounts for 39.8%, while Trip Purpose and Gender contribute smaller but still positive shares. These results indicate

that passenger choice between Korean and foreign carriers is driven mainly by price sensitivity and age related travel behavior.

The decision tree also provides an interpretable set of behavioral rules that explain how passenger characteristics influence airline choice.

The tree produces three clear behavioral rules:

Rule 1: Airfare is the first behavioral divider.

Passengers paying fares below approximately 205,000 KRW (the 20.5 threshold in 10,000 KRW units) behave differently from passengers paying higher fares. Travelers paying lower fares are more likely to select foreign carriers, particularly among older passengers. This suggests that price sensitive travelers are more willing to consider non Korean airlines when searching for cheaper options.

Rule 2: Among low fare passengers, age further differentiates behavior.

Younger low fare passengers under age 25 are associated with foreign carriers, likely reflecting students or younger budget travelers who are more comfortable choosing unfamiliar Asian carriers when prices are lower. Low fare passengers between ages 25 and 45 tend to prefer Korean carriers, while low fare passengers over age 45 again show a higher probability of choosing foreign carriers. This pattern indicates that price sensitivity interacts with age, producing distinct traveler segments within the lower fare market.

Rule 3: Among high fare passengers, Korean carriers dominate across all passenger groups.

Passengers paying above 205,000 KRW are classified primarily as Korean carrier passengers regardless of age, gender, or trip purpose. This pattern is consistent with the premium positioning of Korean Air and Asiana. Once travelers are willing to pay a higher airfare, they overwhelmingly select Korean carriers, indicating strong brand preference in the higher price segment of the market.

Overall, the decision tree highlights an important airfare threshold above which Korean carrier dominance becomes nearly complete. This result suggests that the main competition between Korean and foreign carriers occurs in the lower fare segment, particularly among younger and older travelers. In contrast, the premium segment appears to show strong loyalty toward Korean carriers and relatively limited competition from foreign airlines.

4.6 Integrated Behavioral Summary

Taken together, the four models describe a Seoul airport market with a clear and interpretable behavioral structure.

Airport choice is shaped primarily by trip purpose and age. Business travelers, who account for 66.4 percent of the sample, form the core passenger base for Incheon ICN and are 2.54 times more likely to choose ICN than leisure travelers. Leisure travelers, by contrast, show a clear preference for Gimpo GMP. Age introduces a second layer of segmentation. Passengers aged 55 and above are more likely to favor GMP, and for this group airport choice appears to depend more on the convenience of the ground access journey than on trip purpose alone. In this dataset, gender, seat class, access time, and access cost do not produce statistically significant differences in airport choice, suggesting that the most meaningful behavioral divide is between business and leisure travel and between younger and older passengers.

Airline choice follows a different logic. The measurable demographic and trip variables available in this survey do not significantly distinguish passengers who choose Korean carriers from those who choose foreign carriers. Instead, the decision tree indicates that fare level is the clearest organizing factor. Above an airfare threshold of approximately 205,000 KRW, Korean carriers capture almost all passengers. Below that threshold, competition becomes more active and an interaction between age and fare describes the smaller segment in which foreign carriers remain viable alternatives. The absence of significant demographic predictors suggests that airline loyalty, meaning a habitual preference for specific carriers shaped by prior experience and loyalty program membership, is likely to be the dominant behavioral force, even though this survey was not designed to measure it directly.

These two markets are also closely connected. The business travelers who are most likely to choose ICN are also the passengers most likely to select full service Korean carriers such as Korean Air and Asiana. This pattern is consistent with the behavioral profiles documented by Leon 2011 and Drabas and Wu 2013. Leisure travelers who favor GMP appear to be more price conscious and more willing to consider foreign low cost options, especially on popular short haul routes to Japan, China, and Southeast Asia. Viewed together, the results suggest that airport choice in Seoul is driven mainly by convenience and trip purpose, whereas airline choice is shaped more strongly by fare levels and unobserved loyalty effects.

4.7 Model Evaluation

Model evaluation metrics are reported here for context, not as the primary output They help show the strength of the behavioral patterns captured in the data.

Model	Target	Accuracy	ROC-AUC	Interpretation
Logistic Regression	Airport	64.1%	0.647	Moderate behavioral signal
Decision Tree (depth 3)	Airport	57.7%	—	Supports logistic findings
Logistic Regression	Arline	69.6%	0.574	Weak signal: key variables absent
Decision Tree (depth 3)	Airline	70.4%	—	Fare threshold rule is descriptive

Table 5. Model evaluation summary. Airport naive baseline: 51%. Airline naive baseline: 70% (class imbalance). Accuracy above naive baseline indicates genuine behavioral signal; ROC-AUC above 0.5 confirms discriminative description.

The airport logistic regression captures a meaningful behavioral pattern: its accuracy of 64.1% is higher than the naive baseline of 51%, and its ROC AUC of 0.647 is above the 0.5 threshold for random classification, indicating that trip purpose and age reflect real differences in airport choice behavior. By contrast, the airline logistic model performs much more weakly. Its accuracy of 69.6% is the same as the naive majority class baseline, and its ROC AUC of 0.574 is only slightly above random, suggesting that the demographic variables available in this dataset do not explain airline choice well. The airline decision tree, whose ROC AUC is also close to chance, supports the same conclusion.

5. Conclusions and Recommendations

5.1 Conclusions

This project developed descriptive models of airport and airline choice behavior among 488 passengers in the Seoul Metropolitan Area. The core behavioral findings are summarized below.

- Trip purpose describes airport choice more powerfully than any other measured variable. Business travelers are 2.54 times more likely to choose Incheon (ICN) over Gimpo (GMP) than leisure travelers. This behavioral difference reflects business travelers' prioritization of route connectivity and schedule flexibility, advantages that ICN holds as the larger international hub. This result also replicates the primary finding of Leon (2011) in a North American two airport corridor and reinforces trip purpose as a consistent behavioral driver of airport selection across different metropolitan airport systems.
- Age is the second significant behavioral variable. Each additional year of age reduces the odds of choosing ICN by approximately 2.4 percent. Passengers aged 55 and over are meaningfully more likely to choose GMP. This pattern likely reflects both greater familiarity with GMP as Seoul's historic primary gateway and older passengers' greater sensitivity to access travel time and convenience.
- The airport decision tree reveals a two tier behavioral structure. Age first separates passengers into two groups with different behavioral profiles. Within the younger majority, trip purpose becomes the decisive factor influencing airport selection. For older passengers, airport choice is additionally influenced by access time, creating a more detailed behavioral pattern than the logistic model alone captures.
- Airline choice is not strongly explained by the demographic and trip variables available in this dataset. None of the available demographic or trip variables significantly distinguishes Korean from foreign carrier passengers. This finding likely reflects missing variables that are well documented in the literature, particularly brand loyalty and flight level attributes such as flight frequency and route availability.
- The airline decision tree identifies airfare as the primary behavioral divider in airline choice. Above approximately 205,000 KRW, Korean carriers capture nearly all passengers in the dataset. Below this threshold, price sensitivity interacts with age to create a small

segment where foreign carriers become competitive, particularly among younger and older low fare travelers.

5.2 Quality of Results

The quality of the airport model is moderate and consistent with the limitations of the available data. Two variables reach statistical significance and the model demonstrates meaningful behavioral signal beyond the naive baseline. Compared with studies such as Hess and Polak (2006), which achieved substantially higher model fit, this dataset does not include important flight level attributes such as departure frequency, number of connections, and on time performance. Prior research identifies these attributes as major drivers of airport choice. The current models therefore capture the behavioral effects of passenger characteristics, while a more complete explanation would also incorporate the service level attributes of the airports themselves.

The airline model results remain informative despite the absence of statistically significant coefficients. The results indicate that Korean versus foreign carrier preference in Seoul is not structured by the demographic or trip characteristics observed in this survey. This suggests that other factors, particularly brand loyalty and route level attributes, likely play a larger role in shaping airline choice behavior. Even without statistically significant coefficients, this finding contributes to understanding the structure of airline competition in the market. Future research using datasets that include flight level attributes and loyalty measures would likely provide a more complete explanation of airline choice behavior.

5.3 Policy Recommendations

For Incheon International Airport

- ICN should focus on strengthening its position among business travelers. This can be achieved by improving services that support business travel, such as faster security screening, premium check-in facilities, and increased business lounge capacity. The logistic model result (OR = 2.54 for business travelers) indicates that this group shows the strongest preference for ICN.
- The results also suggest that older passengers may find ICN more complex to navigate. Improving wayfinding systems, clearer signage, and providing assisted navigation services could help make the airport easier to use for older travelers.

- Enhancing transport connectivity between ICN and major business districts in Seoul, such as Gangnam and Yeouido, could further improve accessibility for business passengers.

For Gimpo International Airport

- GMP's primary passenger base consists of leisure travelers and older passengers. Therefore, marketing strategies should highlight the airport's key advantages, such as convenience, proximity to central Seoul, and its relatively simple terminal layout. Since approximately 64% of leisure travelers already prefer GMP, emphasizing these strengths could help reinforce and expand this segment.
- GMP could also consider expanding its international route network, particularly toward short-haul leisure destinations in Japan and Southeast Asia, where passenger demand is relatively high. Introducing additional routes to these destinations would increase the airport's attractiveness to leisure travelers who already prefer GMP but may currently use Incheon International Airport when their desired destinations are not available from Gimpo.

For Airlines Operating in the Seoul MAR

- Korean carriers appear to have a strong presence in the premium fare segment, particularly for tickets priced above 205,000 KRW. The decision tree analysis shows that most passengers purchasing tickets at this price level tend to choose Korean airlines. However, competition is stronger in the lower fare segment, especially among younger passengers (under 25) and older travelers (over 45). To remain competitive, Korean airlines should continue offering price-competitive options on short-haul routes to destinations such as Japan, China, and Southeast Asia, where foreign airlines are particularly active.
- The results also indicate that demographic characteristics do not strongly influence airline choice. This suggests that other factors, such as loyalty programs, may play a more important role in maintaining market share. Airlines should therefore focus on strengthening frequent flyer program enrollment and retention. Previous research by Tomasz Drabas and Chunyan Wu (2013) similarly found that a passenger's previous airline choice is often one of the strongest predictors of their future airline selection.The absence of demographic predictors of airline choice confirms that carrier loyalty programs are

likely the primary behavioral mechanism maintaining market share. Airlines should prioritize frequent flyer program enrollment and retention over demographic targeting, consistent with Drabas and Wu's (2013) finding that past airline choice is the strongest predictor of future selection.

5.4 Limitations and Future Work

Several limitations of the dataset should be acknowledged when interpreting the results of this study. One important limitation relates to missing data in the airport access variables. In particular, access cost and access time contained relatively high levels of missing responses, which reduced the number of observations available for the airport choice analysis. As a result, the effective sample size was smaller than the total survey sample, which may have limited the statistical strength of the results and reduced the ability to fully capture the influence of access-related factors that are commonly identified in previous airport choice studies.

Another limitation is the absence of detailed flight-level service attributes within the dataset. Variables such as flight frequency, connection availability, and on-time performance are widely recognized as important determinants of passenger airport and airline choice. Previous research by Stephane Hess and John W. Polak, as well as by Becky P. Y. Loo, highlights the importance of these service characteristics in shaping traveler decisions. Without these variables, the analysis in this study focuses primarily on passenger and trip characteristics rather than operational service factors.

A further limitation is the lack of information on airline loyalty programs or previous airline usage. Prior studies, including the work of Tomasz Drabas and Chunyan Wu, indicate that past airline choice and loyalty program participation are among the strongest predictors of airline selection. Because such information was not included in the survey, the current analysis relies mainly on demographic and trip variables, which may explain the limited ability of the model to describe airline choice behavior.

Future research could address these limitations in several ways. First, improving the completeness of survey responses, particularly for access-related variables, would strengthen the reliability of airport choice analysis. Second, integrating flight-level information such as service frequency, connection options, and performance indicators would allow for a more comprehensive evaluation of service attributes influencing passenger decisions. Finally, including survey questions related to frequent flyer program membership and previous airline usage would help capture the role of loyalty in airline selection. Expanding the modeling approach to jointly examine airport and airline choice could also provide a more realistic representation of passenger decision-making, as these choices are often made simultaneously rather than independently.

References

Drabas, T., & Wu, C.-L. (2013). Modelling air carrier choices with a Segment Specific Cross Nested Logit model. Journal of Air Transport Management, 32, 8–16.

Hess, S., & Polak, J. W. (2006). Exploring the potential for cross-nesting structures in airport-choice analysis: A case-study of the Greater London area. Transportation Research Part E, 42(2), 63–81.

Leon, S. (2011). Airport choice modeling: Empirical evidence from a non-hub airport. Journal of the Transportation Research Forum, 50(2), 5–16.

Loo, B. P. Y. (2008). Passengers' airport choice within multi-airport regions (MARs): Some insights from a stated preference survey at Hong Kong International Airport. Journal of Transport Geography, 16(2), 117–125.